

**Group Title:** Windows of The World: Evolution of The Windows Operating System

**Group Member Roster:**

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**Group's Topic Theme:**

The group's exhibit theme centers around the rich and specific history of how the Windows OS changed with the ever shifting consumer and industry demands over the past 40+ years, specifically starting when windows was merely an MS-DOS skin or in technical terms a GUI that's on top of an already existing operating system. This theme for the exhibit will also cover the highs and lows of windows, most notable for the "golden age" time circa. 1995 - 2000s was when the Windows OS introduced the 32-bit architecture to the masses as well as being one at the forefront of the plug and play hardware era, in as well as the notable times when the Windows OS hit their noticeable lows circa. 2012 - 2017 when they decided to adapt smartphone OS standards to PC OS standards. Last but not the least this theme will also cover what improvements they are currently making to it or where is the Windows OS heading from here on out.

When it comes to Windows nothing says innovation than a crisis fueled change of which will be also tackled on this exhibit specifically those that involved "Code Red, Blaster, and Slammer" circa. 2001 - 2003, and as well as the windows vista "driver nightmare" 2007, to which the windows OS responded with a lot of improvements based on grounded reality such as the implementation of DEP (Data Execution Prevention): A hardware/software barrier that explicitly marks sectors of memory as "non-executable." If a hacker tries to inject code into a data-only memory slot, the CPU locks up and shuts down the app instantly instead of running the virus, and of course the Windows Driver Framework (WDF), forcing major architectural isolation, to keep the architectural side more compartmentalized and layered against possible crashes in the hardware.

The exhibit will emphasize on the visual and underlying concepts that powers Windows and its evolution from MS-DOS to the modern kernel that is currently utilized by computers and cloud infrastructures. The possible key areas that the group may apply are the MS-DOS shell, WIN32 API, hybrid kernel which uses the Hardware Abstraction Layer and the NTFS file system, and lastly, the modern kernel architecture. The

chronological timeline will showcase the visual differences as well as the underlying concepts involved in the foundation of the different versions of the Windows OS.

**Tech Stack Plan:**

- Node.js
- Astro 6 / Markdown Extended (.mdx)
- React.js
- Tailwind CSS
- Howler.js (for playback of audio elements)

**Interactive Element:**

The interactive element would be a chronological timeline alongside an integrated trivia system that allows users to test their knowledge and earn a score.

The timeline features a series of nodes that maps each iteration of the Windows OS from a simple graphical overlay to a modern cloud-integrated operating system. Smaller regular nodes will be present that highlight key features, breakthroughs, or failures of a specific version of windows while larger major nodes will be used as anchors on the timeline, showcasing a new Windows OS iteration and any revolutionary advancements that were developed alongside it (i.e. The start menu for Windows 95).

Throughout the timeline, interactive checkpoint nodes will be present and clickable. These nodes will quiz the users on what they have currently learned from previous information nodes. Such quizzes may include identifying the Windows OS version by its iconic startup sound, or identifying for what reason a major advancement has been designed.

At the end of the timeline, The user will be shown a tally of their score and a title will be given to them based on their earned score (i.e. Windows expert for 100% score, Windows newcomer for a score <30%)

Design Mockup Snapshot:

